

EXAMPLE 1. Raj can do a piece of work in 20 days and Rohan can do it in 12 days. How long will they take if both work together?

- a. $5\frac{1}{2}$ days b. $7\frac{1}{2}$ days c. $3\frac{1}{2}$ days d. $9\frac{1}{2}$ days

Sol. b. Raj's one day's work = $\frac{1}{20}$

Rohan's one day's work = $\frac{1}{12}$

$$\therefore (\text{Raj} + \text{Rohan})\text{'s one day's work} = \frac{1}{20} + \frac{1}{12} = \frac{3+5}{60} = \frac{8}{60}$$

$\therefore$ Number of day's taken by Raj and Rohan together to complete the work = $\frac{60}{8}$ days or $7\frac{1}{2}$ days

Here, $x = 20$ days and $y = 12$ days

$$\begin{aligned} \therefore \text{Required time} &= \frac{xy}{x+y} \quad [\text{Rule 2}] \\ &= \frac{20 \times 12}{20 + 12} = \frac{20 \times 12}{32} = \frac{15}{2} = 7\frac{1}{2} \text{ days} \end{aligned}$$

EXAMPLE 2. A and B together can do a piece of work in 12 days and A alone can do it 18 days. In how many days can B alone do it?

- a. 14 days b. 24 days c. 36 days d. 28 days

Sol. c. Here, $x = 12$ and $y = 18$

$$\therefore \text{Time taken by B} = \frac{xy}{y-x} = \frac{12 \times 18}{18-12} = 36 \text{ days} \quad [\text{Rule 3}]$$

EXAMPLE 3. A and B can do a piece of work in 3 days. B and C in 9 days and A and C in 12 days. Find the time in which A, B and C can finish the work, working together.

- a. 3 days b. $3\frac{1}{4}$ days
c. $3\frac{15}{19}$ days d. None of these

Sol. c. Here, $x = 3$, $y = 9$ and $z = 12$

According to rule,

$$\begin{aligned} \text{Required time taken by A, B and C} &= \frac{2xyz}{xy + yz + zx} \quad [\text{Rule 4}] \\ &= \frac{2 \times 3 \times 9 \times 12}{3 \times 9 + 9 \times 12 + 3 \times 12} = \frac{6 \times 12}{19} = \frac{72}{19} = 3\frac{15}{19} \text{ days} \end{aligned}$$

(Rule 5) If two groups, M_1 persons of the first group can do ' W_1 ', work in ' D_1 ' days working T_1 h in a day earning a sum of ₹ R_1 and M_2 persons of the second group can do ' W_2 ' work in ' D_2 ' days working T_2 h in a day earning a sum of ₹ R_2 . If each person of both group has the same efficiency of work, then

$$\frac{M_1 D_1 T_1}{W_1 R_1} = \frac{M_2 D_2 T_2}{W_2 R_2}$$

EXAMPLE 4. 15 men complete a work in 16 days. If 24 men are employed, then the time required to complete that work will be

☞ 2014 I

- a. 7 days b. 8 days c. 10 days d. 12 days

Sol. c. Let the work done be 1.

Here, $M_1 = 15$, $D_1 = 16$, $W_1 = W_2 = 1$, $M_2 = 24$ and $D_2 = ?$

Now, according to the formula,

$$\begin{aligned} M_1 D_1 W_2 &= M_2 D_2 W_1 \Rightarrow 15 \times 16 \times 1 = 24 \times D_2 \times 1 \\ \Rightarrow D_2 &= \frac{15 \times 16}{24} = \frac{240}{24} = 10 \text{ days} \end{aligned}$$

Therefore, 10 days are required to complete the work.

(Rule 6) If ' m ' men or ' n ' women can do a piece of work in ' a ' days, then x men and y women can do the same work in $\frac{1}{\frac{x}{m \times a} + \frac{y}{n \times a}}$ days.

EXAMPLE 5. If 3 men or 4 women can reap a field in 43 days. How long will 7 men and 5 women take to reap it?

- a. 3 days b. 7 days c. 12 days d. 15 days

Sol. c. Here, $m = 3$, $n = 4$, $a = 43$, $x = 7$ and $y = 5$

$$\begin{aligned} \therefore \text{Required days} &= \frac{1}{\frac{x}{m \times a} + \frac{y}{n \times a}} = \frac{1}{\frac{7}{3 \times 43} + \frac{5}{4 \times 43}} \quad [\text{Rule 6}] \\ &= \frac{1}{\frac{7+5}{43}} = \frac{1}{\frac{12}{43}} = \frac{43}{12} = 12 \text{ days} \end{aligned}$$

(Rule 7) If A can do a work in x days and B can do $y\%$ fast than A, then B will complete the work in $\frac{100x}{(100+y)}$ days.

EXAMPLE 6. x can do a work in 16 days. In how many days will the work be completed by y , if the efficiency of y is 60% more than that of x ? ☞ 2013 II

- a. 10 days b. 12 days c. 25 days d. 30 days

Sol. a. Here, $x = 16$ days and $y = 60\%$ faster

$$\therefore \text{Required days} = \frac{100x}{100+y} = \frac{100 \times 16}{100+60} = \frac{1600}{160} = 10 \text{ days}$$

[Rule 7]

(Rule 8) If A, B and C can do a piece of work in x , y and z days, respectively and they received ₹ k as wages by working together, then

$$\text{share of A} = ₹ \frac{yz}{xy + yz + zx} \times k$$

$$\text{share of B} = ₹ \frac{xz}{xy + yz + zx} \times k$$

and

$$\text{share of C} = ₹ \frac{xy}{xy + yz + zx} \times k$$

Note If A and B can do a piece of work in x days and y days, respectively and they received ₹ k as wages by working together, then share of $A = ₹ \frac{yk}{x+y}$ and share of $B = ₹ \frac{xk}{x+y}$.

Wages are directly proportional to the work done and indirectly proportional to the time taken by the individual.

EXAMPLE 7. X completes a job in 2 days and Y completes it in 3 days and Z takes 4 days to complete it. If they work together and get ₹ 3900 for the job, then how much amount does Y get?

- a. ₹ 1800 b. ₹ 1200 c. ₹ 900 d. ₹ 800

Sol. b. Here, $a = 2$ days, $b = 3$ days and $c = 4$ days and $k = ₹ 3900$

$$\begin{aligned} \text{Now, amount of } Y &= \frac{ac}{ab+bc+ca} \times k && [\text{Rule 8}] \\ &= \frac{2 \times 4}{2 \times 3 + 3 \times 4 + 4 \times 2} \times 3900 \\ &= \frac{8}{6 + 12 + 8} \times 3900 = \frac{8 \times 3900}{26} = ₹ 1200 \end{aligned}$$

SOME OTHER FORMULAS

1. If A and B can do a piece of work in x and y days, respectively. A and B started working together but A left the work t days before completing the work, then time taken to complete the work will be $\frac{(x+t)y}{(x+y)}$ days.
2. A and B do a piece of work in ' a ' and ' b ' days, respectively. Both begin together but after some days, A leaves the work and the remaining work is completed by B in x days. Then, the time after which A left is given by $T = \frac{(b-x)a}{a+b}$
3. If A and B can do a piece of work in x and y days, respectively. They start working together and after t days B leaves the work, then time taken to finish the whole work will be $\frac{x}{y} \times (y-t)$ days.

EXAMPLE 8. Akshu can do a piece of work in 10 days and Harshal can do same work in 12 days. They started working together but Akshu left the work 2 days before completion of work, then time taken to complete the work?

- a. $6\frac{6}{11}$ days b. $5\frac{3}{10}$ days c. $4\frac{3}{2}$ days d. $7\frac{2}{5}$ days

Sol. a. $6\frac{6}{11}$ days $= \frac{12 \times (10+2)}{10+12} = \frac{12 \times 12}{22} = 6\frac{6}{11}$ days

EXAMPLE 9. A can do a piece of work in 10 days and B can do same work in 15 days. They started working together but after 2 days, A left work. The remaining work was completed by B alone, then time taken to complete the work?

- a. $8\frac{3}{11}$ days b. $8\frac{2}{3}$ days c. $5\frac{3}{4}$ days d. $2\frac{5}{3}$ days

Sol. b. $8\frac{2}{3}$ days $= \frac{10}{15} (15-2) = \frac{26}{3} = 8\frac{2}{3}$ days

PIPES AND CISTERNS

Problems on pipes and cisterns are based on the basic concept of time and work. Pipes are connected to a tank or cistern and are used to fill or empty the tank or cistern.

Important Rules and Formulae

(Rule 9) If there are two pipes A and B takes ' a ' and ' b ' h respectively to fill a tanker, then the two pipes together fill $\left(\frac{1}{a} + \frac{1}{b}\right)$ part of the tank in 1h and time taken to fill the tank will be $\left(\frac{ab}{a+b}\right)$ h. If a pipe fills a tank in ' a ' h, then in 1 h only $\frac{1}{a}$ of the tank is filled.

EXAMPLE 10. Pipe A can fill a tank in 45 h and pipe B can fill it in 36 h. If both the pipes are opened in the empty tank. In how many hours will it be full?

- a. 10 h b. 15 h c. 20 h d. 28 h

Sol. c. Here, $a = 45$ h and $b = 36$ h.

$$\therefore \text{Required time} = \frac{ab}{a+b} = \frac{45 \times 36}{45+36} = \frac{1620}{81} = 20 \text{ h}$$

- (Rule 10)** (i) If a pipe can fill a tank in ' a ' h and another can fill the tank in ' b ' h but a third pipe empties the filled tank in ' c ' h, then in one hour $\left(\frac{1}{a} + \frac{1}{b} - \frac{1}{c}\right)$ part of the tank will be filled (when all the three pipes are open) and time taken to fill the tank will be $\frac{abc}{bc+ac-ab}$ h.
- (ii) If a pipe can fill a tank in a_1 h and another pipe can empty the filled tank in a_2 h, then in one hour $\left(\frac{1}{a_1} - \frac{1}{a_2}\right)$ part of the tank will be filled. (when both pipes are open) and time taken to fill the tank will be $\left(\frac{a_1 a_2}{a_2 - a_1}\right)$ h.

EXAMPLE 11. Two pipes ' A ' and ' B ' can fill a tank in 36 min and 45 min, respectively. A waste pipe ' C ' can empty the tank in 30 min. In how much time the tank is full if all three pipes are opened?

- a. 60 min b. 90 min c. 115 min d. None of these

Sol. a. Here, $a = 36$, $b = 45$ and $c = 30$.

So, time taken to fill the tank will be

$$\begin{aligned} &= \frac{abc}{ac+bc-ab} && [\text{Rule 8}] \\ &= \frac{36 \times 45 \times 30}{36 \times 30 + 45 \times 30 - 36 \times 45} = 60 \text{ min} \end{aligned}$$

> PRACTICE EXERCISE

1. X can do $\frac{3}{4}$ of a work in 12 days. In how many days X can finish the $\frac{1}{2}$ work?
(a) 8 days (b) 16 days (c) 12 days (d) 24 days
2. A can do a piece of work in 10 days and B can do the same work in 12 days. How long will they take to finish the work, if both work together?
(a) $5\frac{5}{11}$ days (b) $6\frac{5}{11}$ days
(c) $5\frac{1}{5}$ days (d) None of these
3. A , B and C working together take 30 min to address a pile of envelopes. A and B together would take 40 min, A and C together would take 45 min. How long would each take working alone?
(a) A : 72 min, B : 90 min, C : 120 min
(b) A : 42 min, B : 90 min, C : 120 min
(c) A : 72 min, B : 90 min, C : 100 min
(d) A : 72 min, B : 80 min, C : 120 min
4. A and B together can do a piece of work in 12 days and A alone can do it in 36 days. In how many days can B alone do it?
(a) 18 days (b) 12 days
(c) 15 days (d) 20 days
5. Ram can do a piece of work in 6 days and Shyam can finish the same work in 12 days. How much work will be finished, if both work together for 2 days?
(a) One-fourth of the work (b) One-third of the work
(c) Half of the work (d) Whole of the work
6. A and B can do given work in 8 days; B and C can do the same work in 12 days and A , B , C complete it in 6 days. In how many days can A and C finish it?
(a) 12 (b) 8 (c) 14 (d) 16
7. A can finish a work in 8 days and B can do it in 12 days. After A had worked for 3 days, B also joins A to finish the remaining work. In how many days will the remaining work be finished?
(a) 2 days (b) 3 days
(c) 4 days (d) 5 days
8. Two taps can fill a tub in 5 min and 7 min, respectively. A pipe can empty it in 3 min. If all the three are kept open simultaneously, when will the tub be full?
(a) 60 min (b) 85 min (c) 90 min (d) 105 min
9. A and B can do a piece of work in 40 days and 50 days, respectively. Both begin together but after a certain time, A leaves off. In this case B finishes the remaining work in 20 days. After how many days did A leave?
(a) 14 days (b) $13\frac{1}{3}$ days (c) 13 days (d) 15 days
10. P and Q can do a job in 2 days, Q and R can do it in 4 days and P and R in $\frac{12}{5}$ days. What is the number of days required for P alone to do the job?
(a) $\frac{5}{2}$ (b) 3 (c) $\frac{14}{5}$ (d) 6
11. $\frac{1}{48}$ of a work is completed in half a day by 5 persons. Then, $\frac{1}{40}$ of the work can be completed by 6 persons in how many days?
(a) 1 (b) 2 (c) 3 (d) $\frac{1}{2}$
12. A garrison of ' n ' men had enough food to last for 30 days. After 10 days, 50 more men joined them. If the food now lasted for 16 days, what is the value of n ?
(a) 200 (b) 240 (c) 280 (d) 320
13. Two pipes A and B can fill a tank in 12 and 16 min, respectively. If both the pipes are opened simultaneously, after how much time should B be closed so that the tank is full in 9 min?
(a) 3 min (b) 5 min (c) 4 min (d) 2 min
14. 2 men undertake to do a job for ₹ 1400. One can do it alone in 7 days and the other in 8 days. With the assistance of a boy they finish the work in 3 days. How should the money be divided?
(a) ₹ 600, ₹ 525, ₹ 275 (b) ₹ 550, ₹ 500, ₹ 350
(c) ₹ 650, ₹ 470, ₹ 280 (d) None of these
15. A , B and C can do a piece of work individually in 8, 10 and 15 days, respectively. A and B start working but A quits after working for 2 days. After this, C joins B till the completion of work. In how many days will the work be completed?
(a) $\frac{53}{9}$ days (b) $\frac{34}{7}$ days
(c) $\frac{85}{13}$ days (d) $\frac{53}{10}$ days
16. A can do a piece of work in ' x ' days and B can do the same work $3x$ days. To finish the work together they take 12 days. What is the value of ' x '?
(a) 8 (b) 10 (c) 12 (d) 16

- 17.** If 6 men and 8 boys can do a piece of work in 10 days while 26 men and 48 boys can do the same in 2 days, what is the time taken by 15 men and 20 boys in doing the same type of work?
(a) 4 days (b) 5 days (c) 6 days (d) 7 days
- 18.** Four taps can individually fill a cistern of water in 1h, 2h, 3h and 6h, respectively. If all the four taps are opened simultaneously, the cistern can be filled in how many minutes?
(a) 20 (b) 30 (c) 35 (d) 40
- 19.** 76 ladies complete a job in 33 days. Due to some reason some ladies did not join the work and therefore, it was completed in 44 days. The number of ladies who did not report for the work is
(a) 17 (b) 18 (c) 19 (d) 20
- 20.** 9 men finish one-third work in 10 days. The number of additional men required for finishing the remaining work in 2 more days will be
(a) 78 (b) 81 (c) 55 (d) 30
- 21.** Ravi and Sneha working separately can finish a job in 8 and 12 h, respectively. If they work for an hour alternately, Ravi beginning at 9:00 am. When will the job be finished?
(a) 7 : 30 pm (b) 7 : 00 pm (c) 6 : 30 pm (d) 6 : 00 pm
- 22.** Consider the following statements:
I. If 18 men can earn ₹ 1440 in 5 days, then 10 men can earn ₹ 1280 in 6 days.
II. If 16 men can earn ₹ 1120 in 7 days, then 21 men can earn ₹ 800 in 4 days.
Which of the statement(s) given above is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

Directions (Q. Nos. 23-24) If a_1 men and b_1 boys can complete a work in x days, while a_2 men and b_2 boys can complete the same work in y days, then

$$\frac{\text{One day work of 1 man}}{\text{One day work of 1 boy}} = \frac{(yb_2 - xb_1)}{(xa_1 - ya_2)}$$

- 23.** If 14 men and 12 boys can finish a work in 4 days, while 8 men and 16 boys can finish the same work in 5 days. Compare the 1 day work of 1 man and 1 boy.
(a) 2 (b) $1\frac{1}{2}$ (c) $\frac{1}{2}$ (d) 3
- 24.** 28 men and ' m ' boys can finish a work in 4 days, while 20 men and 6 boys can finish the same work in 5 days. Now, if the ratio of 1 day work of 1 man and 1 boy is 1 : 2, what will be the value of ' m '?
(a) 7 (b) 6 (c) 8 (d) 12
- 25.** 45 people take 18 days to dig a pond. If the pond would have to be dig in 15 days, then the number of people to be employed will be **2012 I**
(a) 50 (b) 54 (c) 60 (d) 72
- 26.** A and B can do a piece of work in 10 h. B and C can do it in 15 h, while A and C take 12 h to complete the work. B independently can complete the work in **2012 I**
(a) 12 h (b) 16 h (c) 20 h (d) 24 h
- 27.** X can do a piece of work in 25 days. Y is 25% more efficient than X. The number of days taken by Y is **2012 II**
(a) 15 days (b) 20 days (c) 21 days (d) 30 days
- 28.** A mason can build a tank in 12 h. After working for 6 h, he took the help of a boy and finished the work in another 5 h. The time that the boy will take alone to complete the work is **2013 I**
(a) 30 h (b) 45 h (c) 60 h (d) 64 h
- 29.** X can complete a job in 12 days. If X and Y work together, they can complete the job in $6\frac{2}{3}$ days. Y alone can complete the job in **2013 I**
(a) 10 days (b) 12 days (c) 15 days (d) 18 days
- 30.** Pipe A can fill a tank in 10 min and pipe B can empty it in 15 min. If both the pipes are opened in an empty tank, the time taken to make it full is **2013 I**
(a) 20 min (b) 25 min
(c) 30 min (d) None of these
- 31.** 4 goats and 6 sheeps can graze a field in 50 days. 2 goats and 9 sheeps can graze the field in **2013 II**
(a) 100 days (b) 75 days (c) 50 days (d) 25 days
- 32.** A can finish a work in 15 days, B in 20 days and C in 25 days. All these three worked together and earned ₹ 4700. The share of C is **2013 II**
(a) ₹ 1200 (b) ₹ 1500 (c) ₹ 1800 (d) ₹ 2000
- 33.** 20 workers working for 5 h per day complete a work in 10 days. If 25 workers are employed to work 10 h per day, what is the time required to complete the work? **2013 II**
(a) 4 days (b) 5 days (c) 6 days (d) 8 days
- 34.** 18 men can earn ₹ 360 in 5 days. How much money will 15 men earn in 9 days? **2013 II**
(a) ₹ 600 (b) ₹ 540 (c) ₹ 480 (d) ₹ 360
- 35.** 2 men and 1 woman can complete a piece of work in 14 days, while 4 women and 2 men can do the same work in 8 days. If a man gets ₹ 90 per day, what should be the wages per day of a woman? **2013 II**
(a) ₹ 48 (b) ₹ 60 (c) ₹ 72 (d) ₹ 135

PREVIOUS YEARS' QUESTIONS

- 36.** A can do a piece of work in 4 days and B can complete the same work in 12 days. What is the number of days required to do the same work together? **2013 II**
(a) 2 days (b) 3 days (c) 4 days (d) 5 days
- 37.** A , B and C can do a piece of work individually in 8, 12 and 15 days, respectively. A and B start working but A quits after working for 2 days. After this, C joins B till the completion of work. In how many days will the work be completed? **2014 II**
(a) $5\frac{8}{9}$ days (b) $4\frac{6}{7}$ days (c) $6\frac{7}{13}$ days (d) $3\frac{3}{4}$ days
- 38.** A is thrice as efficient as B and hence completes a work in 40 days less than the number of days taken by B . What will be the number of days taken by both of them when working together? **2014 II**
(a) 22.5 days (b) 15 days (c) 20 days (d) 18 days
- 39.** The efficiency of P is twice that of Q , whereas the efficiency of P and Q together is three times that of R . If P , Q and R work together on a job, in what ratio should they share their earnings? **2015 I**
(a) 2 : 1 : 1 (b) 4 : 2 : 1 (c) 4 : 3 : 2 (d) 4 : 2 : 3
- 40.** A and B are two taps which can fill a tank individually in 10 min and 20 min, respectively. However, there is a leakage at the bottom, which can empty a filled tank in 40 min. If the tank is empty initially, then how much time will both the taps take to fill the tank with leakage? **2015 II**
(a) 2 min (b) 4 min (c) 5 min (d) 8 min
- 41.** If 4 men working 4 h per day for 4 days complete 4 units of work, then how many units of work will be completed by 2 men working for 2 h per day in 2 days? **2015 II**
(a) 2 (b) 1 (c) $\frac{1}{2}$ (d) $\frac{1}{8}$
- 42.** If m persons can paint a house in d days, then how many days will it take for $(m + 2)$ persons to paint the same house? **2015 II**
(a) $md + 2$ (b) $md - 2$ (c) $\frac{m+2}{md}$ (d) $\frac{md}{m+2}$
- 43.** Two pipes A and B can fill a tank in 60 min and 75 min, respectively. There is also an outlet C . If A , B and C are opened together, then the tank is full in 50 min. How much time will be taken by C to empty the full tank? **2016 I**
(a) 100 min (b) 110 min (c) 120 min (d) 125 min

ANSWERS

1	a	2	a	3	a	4	a	5	c	6	b	7	b	8	d	9	b	10	b
11	d	12	a	13	c	14	a	15	d	16	d	17	a	18	b	19	c	20	b
21	c	22	d	23	a	24	b	25	b	26	d	27	b	28	c	29	c	30	c
31	d	32	a	33	a	34	b	35	b	36	b	37	a	38	b	39	a	40	d
41	c	42	d	43	a														

HINTS AND SOLUTIONS

- 1.** (a) Since, X can do $\frac{3}{4}$ of work in 12 days.
[Rule 1]
So, X can do 1 work in $\frac{12 \times 4}{3}$ days.
 $\therefore X$ can do $\frac{1}{2}$ work in $\frac{12 \times 4 \times 1}{3 \times 2} = 8$ days
- 2.** (a) Here, $x = 10$ and $y = 12$
 $\therefore$ Number of days taken by A and B
 $= \frac{xy}{x+y} = \frac{12 \times 10}{12+10}$ [Rule 2]

$$= 5\frac{5}{11} \text{ days}$$

- 3.** (a) B 's one min's work
 $= (A + B + C)$'s one min's work
 $-(A + C)$'s one min's work
 $= \frac{1}{30} - \frac{1}{45} = \frac{6-4}{180} = \frac{2}{180} = \frac{1}{90}$
 C 's one min's work $= (A + B + C)$'s one min work $-(A + B)$'s one min work

$$= \frac{1}{30} - \frac{1}{40} = \frac{4-3}{120} = \frac{1}{120}$$

A 's one min's work $= (A + B)$'s one min work $- B$'s one min work
 $= \frac{1}{40} - \frac{1}{90} = \frac{9-4}{360} = \frac{5}{360} = \frac{1}{72}$
Hence, A , B and C alone can finish the work in 72 min, 90 min and 120 min, respectively.